Time-Series-Based Stock Market Analysis using Machine Learning

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***Abstract*—The stock market is a very complex and volatile entity. Predicting it is a very challenging task but the emergence of Machine Learning and the availability of large-scale stock market data provide an opportunity to leverage data-driven approaches for improving prediction accuracy. In this paper, a framework for stock market analysis using up-to-date financial data extracted from the Yahoo Finance library and machine learning models is proposed. This paper's approach is to capture the complex relationships and patterns in the data by using machine learning techniques. Specifically, machine learning algorithms like XGBoost, Gradient Boost, and Random Forests are employed. By analyzing diverse data sources from various companies, the aim is to extract valuable insights for stock prediction. The experiments involve processing contemporary stock data, implementing machine learning models on it, and comparing the outcomes. The predominant motives are to enhance stock prediction accuracy, manage inherent risks, enable real-time decision-making, and maximize profitability by providing valuable insights for investors that help optimize their portfolios. However, the results are real-time and the optimal algorithm for a particular stock is decided on a dynamic basis due to the stock market being volatile and unpredictable in nature.**

***Index Terms*—Stock Market Analysis, Machine Learning, Yahoo Finance, XGBoost, Gradient Boost, Decision Tree, Random Forest**

I. Introduction

Stock market plays a very crucial role in the global economy [1] [2] [3]. It provides a place to trade securities, stocks, bonds, and derivatives. Investors and financial professionals rely on stock market analysis to make informed decisions and maximize their returns. However, accurately predicting stock prices is a complex task because of various factors such as market dynamics, economic indicators, investor sentiments, and external events.

The vast data generated in the stock market presents both opportunities and challenges for stock forecasting. As online trading platforms, social media, and financial news outlets have mushroomed in number the amount and range of data involved has multiplied. These data items include historical prices, company financials, market news, and social media sentiments. Harnessing the power of Yahoo Finance (yfinance) a library that provides live stock market data offers a way to effectively process, analyse and derive insights from stock data.

Typical traditional data analysis techniques [4] [5] [6] [7] may miss hidden patterns and correlations, which is one of the major advantages of yfinance in stock prediction. In various fields like finance, traditional data analysis approaches like statistical models and basic regression models are considered essential for analysing data. However, they often fail when it comes to capturing intricate and non-linear relationships present in market data. To learn complex relationships and make predictions based on historical patterns Machine learning algorithms can be trained on large datasets. These algorithms uncover intricate connections and patterns that conventional approaches can miss.

By leveraging machine learning, stock market participants can gain deeper insights into market trends, identify investment opportunities, and make more informed decisions. It is important to address challenges like data quality, feature engineering, model interpretability, and the robust nature of financial markets to ensure reliable and robust predictions. The combination of real-time stock data and advanced machine learning techniques offers an effective approach to tackle the complexities of stock market prediction. By using the power of real-time data, computational scalability, and sophisticated algorithms, this paper aims to develop a robust system that empowers investors and financial professionals with accurate predictions and valuable insights to navigate the competitive stock market landscape.

The research will begin with conducting an extensive review of relevant literature in section II, focusing on major publications and prominent academic articles in the field. An in-depth analysis of their research methods and outcomes will be conducted. Next, an in-detail description of the dataset employed in this paper will be presented in section III elucidating its composition and its relevance to the research. Subsequently, the methodology in section IV will explain the methodologies and strategies that are being used. Section V will be a results section that will present the outcomes of the analysis conducted which will offer a clear understanding of the implications of this work. Lastly, a conclusion and a references section as section VI will describe the summary of the work and acknowledge the fundamental works that have shaped and impacted this study.

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The key contributions of the proposed system are:

- The model experiments with various machine learning models which include SVM, Random Forest, XGBoost, and Gradient Boost for stock market prediction tasks.
- The utilization of time series stock market data to anticipate market movements using real-time analysis aiming to enhance accuracy and reliability in forecasting market trends.

II. Related Works

Delving into a diverse array of stock market topics explored in twelve papers. These scholarly works help gain valuable insights into the evolving landscape of stock market analysis. From studying versatile data and analysing it to being able to predict the price and evaluate the stock performance and other measures, each paper offers a unique perspective on the challenges faced in the world of data analysis. Together, these provide an overview of the cutting-edge techniques, methods, and approaches employed in understanding, analysing and predicting the trends of stock market data. Join us in exploring the complexities of these study endeavors, uncovering the innovations that contribute to the continuous attempts to comprehend and forecast the stock market.

Ashok et al. in her research discusses the importance of stock trading [8] and the growing interest in using machine learning methods to predict stock prices. The study highlights the use of the auto-regressive integrated moving average (ARIMA) for analysis in stock price prediction. The research parallelly focuses on using LSTM networks to predict stock movements on basis of past data. Experimental results show that ARIMA has a strong potential for short-term predictions and competes with other existing techniques. The results indicate that the LSTM model achieved best performance, accurately projecting the market with a precision of 77%.

The work by Binoy et al. presents an optimized decision tree-SVM hybrid genetic algorithm system [9] for stock market forecasting which outperforms other systems such as artificial neural networks and naive Bayes-based systems in terms of prediction accuracy. The system helps transform the task to classification which improves the accuracy. The system performance is validated using time series data from the BSE-Sensex. It is also mentioned in the paper that imperfect characteristics of the stock market make it challenging to develop accurate prediction systems. A decision tree-SVM hybrid optimized using a genetic algorithm allows the system to adapt to changing market conditions.

In their research on predicting the stock market using improved LSTM Preethi et al. [10] predicted prices in the context of three tire companies: MRF Limited, CEAT Limited, and JK Tire Industries Limited. Their proposed LSTM model incorporates useful values such as RSI and SMA for training, aiming to improve prediction accuracy. The paper uses metrices such as MAPE, MSE, Precision, and Recall to evaluate the performance of the LSTM model. The results demonstrated that the model accuracies of opening prices are 96%, 95%, and 95% respectively, and that of the closing prices are 95%, 95%, and 93% respectively.

The findings by Arul et al. showcased the utilization of attention mechanisms and decomposition based on variational modes [11] for forecasting intraday stock prices. The study uses variational mode decomposition (VMD) to disseminate the stock data into different processes and then trains the long short-term temporal (LSTM) network with an attention mechanism on it. Experimental results demonstrate the effectiveness of VMD it gave a mean absolute error (MAE) score of 163.91 and LSTM with VMD achieved a root mean square error (RMSE) score of 192.39. In addition, the inclusion of attention mechanism led to improvement in the prediction, resulting in an MAE score of 94.16 and a 117.12 score for RMSE for the VMD + LSTM + attention model. Future study is advised to explore alternative decomposition approaches, and compare with other LSTM variants and transformer applications.

Singh et al. proposed a real-time stock prediction strategy [12] for intraday trading. This paper advocates for two approaches: Incremental and Offline – Online learning. The study applies these approaches to univariate and multivariate time-series data. Experimental evaluations were conducted on eight liquid stocks from NASDAQ and NSE exchanges and it is demonstrated that Offline–Online has a lower forecasting error when compared to incremental models. Models are assessed using metrics such as MAE, RMSE, and Mean Absolute Percentage Error (MAPE). Multivariate models, particularly Bi-LSTM, exhibit better accuracy in predicting stock prices, offering potential benefits for short-term trading strategies in real-time scenarios. Moreover, the paper suggests the inclusion of deep learning with incremental approaches. Therefore, Ishwarappa et al. presented an approach [13] that uses a Deep Reinforcement Long Short-Term Memory (DRLSTM) model for stock price forecasting based on technical indicators. The model is evaluated based on a dataset spanning 900 days (2000 to 2020) from the NIFTY 50. Correlation analysis and feature extraction are utilized in this study. The focus is on OHLC (Open, High, Low, Close) technical variables to analyze three banking sector stocks: SBI, HDFC, and BajajFin. The results showed that the proposed DRLSTM model achieved a low Mean Squared Error (MSE) value of 0.017% for SBI, 0.014% for HDFC, and 0.018% for BajajFin indicating that this approach outperforms other techniques in terms of accuracy, achieving over 97% accuracy in predicting stock trends for the three banking sectors.

In the study proposed by Hind et.al big data analytics techniques [14] specifically Spark, Hive, and MapReduce were applied to analyze the stock performance of Al Rajhi Bank. They aim to investigate the five-month peak of Alrajhi shares in 2022 using MapReduce, track the five lowest prices at which the market started to develop in 2022 using Spark programming and track the highest volumes in the last five years using the Hive method. This study highlights the effectiveness of the MapReduce programming model in processing unstructured data, providing insights into the Al Rajhi stock trends and encouraging further exploration of techniques for processing

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such data.

Rouf and team. provided a comprehensive overview of stock market prediction approaches [15] from the last decade (2011–2021). The authors emphasized the integration of ML, big data analytics, and DL into stock market prediction frameworks. The comparative analysis indicates that Support Vector Machine (SVM) is a favorable choice, while techniques like Artificial Neural Networks (ANN) and Deep Neural Networks (DNN) are gaining prominence for their accuracy and efficiency. Notably, the article highlights the significance of considering social media sentiments, and cyber threats in contemporary stock markets. Both market data and textual data are included, this serves as the key factor in improving prediction accuracies. Another such approach experimented by Bansal and team explored the implementation of machine learning techniques [16] for stock market prediction, specifically focusing on five specific algorithms: KNN, LR, SVM, Decision Tree, and LSTM. The research work includes an extensive data implementation using a dataset consisting of the prices of 12 major companies in the Indian stock market over the last 7 years. A comparative analysis using metrics like Symmetric Mean Absolute Percentage Error (SMAPE), R2, and RMSE is performed. The conclusion of this research work highlights that the Long Short-Term Memory (LSTM) algorithm outperforms other algorithms with extensively accurate results for stock price prediction in the market.

An enhanced version of the above research is provided by Pragathi et al., who with her team proposed a real-time stock prediction framework using deep reinforcement learning [17]. They specifically focused on reducing overestimation in Artificial Neural Network (ANN) models caused due to market volatility. The framework employs Transformed Deep-Q Learning to model stock data and technical indicators, such as Estimated Moving Average (EMA), Relative Strength Index (RSI), and Accumulation Distribution Indicator (ADO). The architecture is designed for cost reduction and transaction time optimization, utilizing GPU and memory containers for real-time predictions. The stock predictor is implemented with a GRPC-based clean architecture, allowing easy updates, and new service additions with lesser integration costs. The feature of the cloud to archive data contributes to cost reduction. The framework is tested on real-time stock data and results indicate improved profit percentages compared to traditional Actor-Critic Deep-Q models. Some future enhancements include integrating sentiment analysis and exploration of FPGA containers for further real-time performance optimization.

The work by Javed and team. explores the application of big data analytics [18] for predicting stock market trends using social media data and machine learning models. The study focuses on leveraging Spark MLlib through PySpark for scalability, speed, and integration with other tools. The authors implemented various models, including LR, Generalized Linear Regression (GLR), logistic regression, Random Forest (RF), Decision Tree, and Naive Bayes (NB) to analyze the historical stock prices of 10 top companies. Additionally, sentiment analysis from social media platforms like Twitter and Yahoo Finance was incorporated to enhance predictions. Experiments showed that generalized linear regression, random forest, and linear regression achieved accuracy rates of 80% to 98% in predicting stock price movements. However, the decision tree model did not perform well in this regard. The study also recommends exploring deep learning models such as LSTM for future research.

The research by Zhihao Peng et al. put forward a robust data pipeline based on a Cloudera-Hadoop blended model [19] for analyzing and predicting the daily gains of a few shortlisted US stocks based on real-time data from Yahoo Finance. US oil stocks, particularly those from the US Oil Fund (USO), are used as an example dataset. The pipeline includes steps such as data storage, data acquisition, data characterization, data injection into Hadoop Distributed File System (HDFS), pre-processing using Pyspark, and machine learning with Mlib functions in Spark. Results indicate that linear regression describes 3% of USO stock price variations with a 1.95% Mean Average Error. The study aims to assist in identifying potentially profitable stocks for trading and suggests further exploration of improvements, including automated analysis processes and testing Neural Network models for more accurate predictions of US stock prices.

In a lump sum, this section provides an overview of various approaches employed in stock market analysis and prediction. The literature survey talks about data collection and preprocessing in various ways from datasets to pipeline models. The significance of Machine Learning models in stock market forecasting is highlighted. Subsequently, the study of Deep Learning models like LSTM, ANN, and big data approaches through Pyspark also gives insight into their applications in stock market analysis. These works reinforce the capability of ML in improving the accuracy and efficiency of stock analysis and highlight the role of big data analytic platforms in contributing towards this goal. These valuable insights serve as catalysts for ongoing research and advancement into the intricate world of stock market forecasting propelling towards even more resilient and efficient investment tactics in the constantly evolving financial sphere.

This study will focus on extracting real-time stock market data and processing it using Machine Learning models to finally advise the investor about the buyability decision of the stock. The system would first extract real-time data through a module and then execute ML on the data. After execution, the models are evaluated using Accuracy and F1-Score to assess their performance.

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III. Dataset Description

To create a comprehensive stock market analysis-cum-recommendation system, it is crucial to have a wide-ranging dataset that encompasses historical stock price data, trading volume, and relevant market indicators. The dataset is collected through a library called YFinance. Yahoo Finance (yfinance) is a widely recognized and reliable platform for financial information and the data is both validated and recent. It provides us with the previous stock data of various companies along with a detailed look at the performance of the company's index and its constituent stocks. The extracted data is adaptable and seamlessly dynamic. Users can tailor the data to meet their specific needs by modifying their inputs. The primary purpose is to analyze various trends in the stock market and the factors that are affecting it. Given below is the list of a few features and derived attributes extracted from the dataset. In this approach, the data is extracted for the past 5 years from the yfinance library and stored in a CSV file. The size of this dataset is 1300 rows.

Features: The dataset has many features namely: Date, Symbol, Series, High (highest price of the day), Prev Close (previous day's closing price), Open (open price of the day), Last (last traded price in that day), Low (lowest price in the day), Close (close price of the day), VWAP (volume weighted average price), Volume and Turnover.

Derived Attributes: Some attributes were additionally derived like Simple Moving Average (SMA 50 and SMA 200), Signal (Decision Factor), Dividends, Stock Splits, Volatility, Daily Return, Price Change, High Low Difference, Volume Change

Date	Open	High	Low	Close	Adj Close	Volume
24-12-2018	123.1	129.74	123.02	124.06	124.06	22066000
26-12-2018	126	134.24	125.89	134.18	134.18	39723400
27-12-2018	132.44	134.99	129.67	134.52	134.52	31202500
28-12-2018	135.34	135.92	132.2	133.2	133.2	22627600
31-12-2018	134.45	134.64	129.95	131.09	131.09	24625300
02-01-2019	128.99	137.51	128.56	135.68	135.68	28146200
03-01-2019	134.69	137.17	131.12	131.74	131.74	22717900
04-01-2019	134.01	138	133.75	137.95	137.95	29002100
07-01-2019	137.56	138.87	135.91	138.05	138.05	20089300
08-01-2019	139.89	143.14	139.54	142.53	142.53	26263800
09-01-2019	142.95	144.7	141.27	144.23	144.23	22205900
10-01-2019	143.08	144.56	140.84	144.2	144.2	16125000
11-01-2019	143.15	145.36	142.57	143.8	143.8	12908000
14-01-2019	142	146.57	141.27	145.39	145.39	20520300
15-01-2019	146.01	150.68	145.99	148.95	148.95	24069000
16-01-2019	149	149.65	147	147.54	147.54	18025700
17-01-2019	146.95	149	146.5	148.3	148.3	15787900
18-01-2019	149.75	152.43	148.55	150.04	150.04	31029600

Fig. 1: Primary Dataset

The primary dataset extracted from yfinance is illustrated in Fig.1 it encapsulates crucial stock market data with features like date, volume, open, low, close, and high. The most prominent and useful attributes that are used in this research are displayed in Fig.1. This dataset serves as the cornerstone for analyzing the market trends in this paper.

Dividends	Stock Split	SMA_50	SMA_200	Daily_Ret	Volatility	Price_Cha	High_Low	Volume_C	Signal
0	0	184.308	176.4051	0.011814	0.016314	0.690002	2.779999	-0.20176	1
0	0	184.024	176.685	0.001001	0.016217	-0.29001	2.440002	0.138694	1
0	0	183.8532	176.935	0.023107	0.016553	2.040009	4.350006	0.735096	1
0	0	183.7384	177.1788	-0.00494	0.01635	-0.92	2.049988	-0.53076	1
0	0	183.8816	177.4573	0.030609	0.015984	5.089996	6.720001	1.146603	1
0	0	183.9824	177.7496	0.003494	0.015854	1.229996	2.800003	-0.3007	1
0	0	184.0872	178.0231	0.004432	0.015858	0.089996	2.069992	-0.0869	1
0	0	184.001	178.2937	-0.02385	0.015764	-4.37999	7.570007	0.75718	1
0	0	184.0392	178.5527	0.021038	0.015954	2.720001	3.160004	-0.51864	1
0	0	183.9786	178.7742	-0.0391	0.0168	-7.66	9.149994	1.405338	1
0	0	183.9326	178.9923	0.020895	0.016898	4.139999	4.380005	-0.37042	1
0	0	184.066	179.203	0.001236	0.015523	1.76001	3.929993	-0.0721	1
0	0	184.172	179.4215	0.008102	0.015405	2.059998	3.910004	-0.29371	1
0	0	184.286	179.6495	0.008037	0.015421	2.199997	4.449997	0.694276	1
0	0	184.3488	179.8691	-0.00048	0.015312	-2.38	4.059998	-0.00608	1
0	0	184.4376	180.0656	-0.0056	0.015221	-1.31	4.470001	1.116745	1
0	0	184.5996	180.2861	0.018061	0.015416	-5.05	9.839996	0.471623	1
0	0	184.8312	180.5127	0.010279	0.015413	0.769989	4.199997	-0.48655	1

Fig. 2: Auxiliary Additions

In addition to the core attributes some supplementary features as illustrated in Fig.2 such as dividends, stock splits, 50-day simple moving average (SMA50), 200-day simple moving average (SMA200), volatility metrics, signal, etc. were added. Fig.2 exemplifies these auxiliary additions. Dividends reflect the company's profitability and stock splits help improve share liquidity. The SMA50 and SMA200 reveal short-term and long-term trends respectively. Volatility metrics indicate price fluctuation and risk, and signal indicators help predict future price movements. These augmented additions along with primary features provide a richer context for market analysis and offer deeper insights for strategic decision-making in the stock market.

IV. Proposed Methodology

In response to traditional challenges, this paper introduces a data-driven and machine learning-centric methodology for enhancing the accuracy and in-depth analysis of the stock market. The data is collected in real-time through the help of the Yahoo Finance library, encompassing attributes like open, high, low, close, and volume, these symbolize the characteristics of stocks. Additionally, extra attributes derived from the primary attributes are appended. These attributes help in better reckoning, analysis, and the estimation of the stock variability with respect to the market.

The integration of Machine learning models further elevates the approach. By leveraging algorithms like SVM, Na¨ıve Bayes, Logistic regression, Random Forest, XGBoost, and Gradient Boost models, the intricate nature of the stocks is aimed to be unraveled. The ultimate objective is to arrive at well-informed conclusions and furnish investors with dependable insights to facilitate better investment decisions. Utilizing the capabilities of the yfinance library, the focus is on extracting records pertaining to a specific company by utilizing its unique stock symbol. The extracted data contains the stock price parameters of the company. This approach helps us streamline the data collection process, therefore allowing for a more effective exploration of the chosen company's financial landscape.

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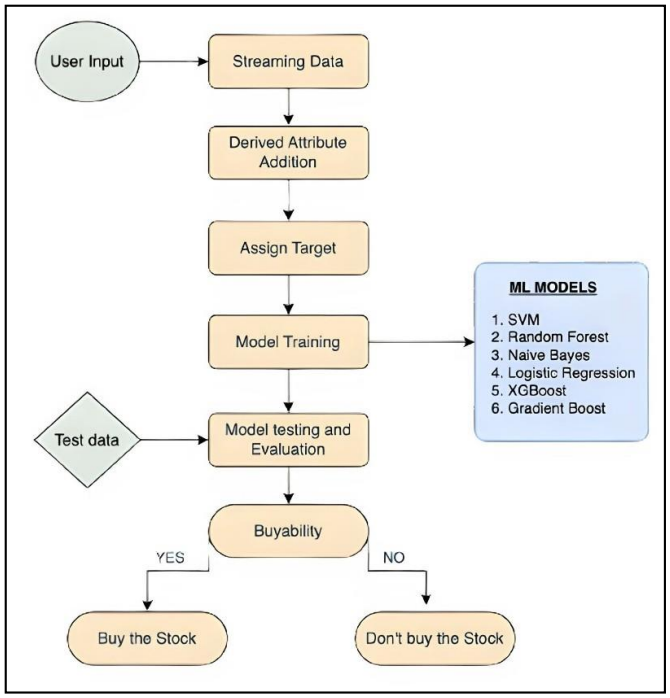

Fig. 3: System Design

The flowchart depicted in Fig.3 illustrates the detailed process in a thorough manner. It follows a logical flow it starts with gathering historical data based and driven by the user input and preparing it for analysis. The data is then extracted and streamed onto the platform. Then derived attributes are added to it and are assigned a target value. Once this is done the model selection and training phase takes place. Here you first choose a machine learning model for example the Support Vector Machine (SVM) and train it on the prepared data. The model is then used to make predictions which are evaluated for accuracy and compared mutually to determine the finest. Finally, a recommendation is provided to the individual regarding whether to buy the stock or refrain from doing so. This flowchart provides a general overview of the approach on leveraging real-time data to forecast prices.

Throughout this research journey, the emphasis will be on how data processing, attribute derivation, model training, and evaluation metrics combine to form a powerful framework for building the model.

A. Streaming Data and Attribute Addition:

A set of carefully crafted derived attributes is introduced to enhance the depth of stock market analysis. These attributes contribute to a holistic understanding of stock behavior. These additions are designed to capture the complex aspects of stock behavior beyond the conventional attributes.

Simple Moving Averages and Signal are the most prominent attributes augmented. SMA 50 represents the average price over the last 50 days, while SMA 200 spans a period of 200 days. Beyond moving averages and signals some additional attributes such as Dividends, Stock Splits, Volatility, Daily Return, Price Change, High Low Difference, and Volume Change are added. These derived attributes play a key role in enhancing the dataset, making it possible to examine stock patterns in greater detail and supporting accurate forecasts made by machine learning models.

B. Assigning Target:

The target attribute represents the signal derived from the interplay between the moving averages and is important in identifying the potential shifts in market sentiment. The Signal attribute is derived from the comparison of SMA 50 with SMA 200. If the SMA 50 is greater than SMA 200, then we assign 1, and if it is lesser than we assign 0. It acts as a pivotal decision factor: a Signal of 1 signifies a favorable buying opportunity (a potential bullish trend), while 0 advises caution (a potential bearish trend).

C. Model Training:

Once the derived attributes are incorporated into the data, a diverse set of machine learning algorithms are applied. Each machine-learning model possesses unique characteristics that are beneficial in their unique ways. A few models like SVM, Naive Bayes, Logistic Regression, XGBoost, and Gradient Boost are deployed. The trained models incorporate diverse attributes, including Simple Moving Averages, Signal indicators and other derived features. The ultimate objective is to arrive at well-informed conclusions regarding the decision to buy or abstain from buying a particular stock. By harnessing the power of machine learning, the methodology aims to provide investors with reliable insights, helping them make better data-driven decisions.

D. Model Evaluation Metrices:

Post-training of the model, some tools are required to help us assess how well the model is performing compared to other models. For this requirement two evaluation metrics are utilized: Accuracy and F1 score. Accuracy can be calculated as a ratio of the number of observations correctly classified to the total number of observations. It is used to provide an average estimate of accuracy in the categorization problems. The F1 score represents the harmonic mean of the values obtained using the precision and the recall parameters. It combines accuracy as the ratio of true positives to those that are positive in prediction and recall as the ratio of true positives to actual positives. It is beneficial when there is an unequal class distribution. These metrics are the benchmarks for evaluating the reliability of the models.

Essentially, this assessment includes the primary scoring fundamental factor Accuracy- The ratio of correctly classified instances among total instances which provides a measure of the complete model's correctness. By assessing the accuracy, insights into the effectiveness of the models across various

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dimensions are gained which subsequently guarantees a thorough assessment of their predictive power.

E. Buyability Parameter:

After models are evaluated on the metrices, an optimal model (the one with the highest metrics) is obtained for that stock. This model will be used to determine whether the stock can be bought or not based on the comparison of SMA 50 with SMA 200. The buyability parameter is decided by the target attribute prediction of the model, if it gives a value of 1 it signifies a favorable buying opportunity indicating a potential bullish trend. Conversely, if the predicted target value is 0 it suggests caution and signals a bearish trend. This assessment provides valuable insights into market sentiments, thereby allowing investors to make informed data-driven decisions.

V. Results

The results of this analysis indicate that determining a singular best model proves challenging because the data is real-time based and it can change over time. The evaluation criteria are based on two metrics: Accuracy and F1 measure. The optimal model is identified based on an interconnected consideration, where higher accuracy and higher F1 score are pivotal. Throughout the experiments conducted even though no model proves to be the best fit, XGBoost and Gradient Boost are strong contenders, showcasing competitive performance in most instances. The fact that there is a lack of a universally applicable solution highlights how difficult forecasting of the stock market is. Variability in factors like data patterns and market dynamics may sometimes lead to instances where other models such as SVM, Naive Bayes, Logistic Regression, or Linear Regression, outperform XGBoost and Gradient Boost. The dynamic nature of financial markets underlines the significance of an adaptable strategy, understanding that no single model universally excels in all conditions.

The experimental findings reveal that all implemented models were able to achieve satisfactory results on stock data. Based on their respective performances across various metrics evaluated in this study, it cannot be confidently asserted that a particular model is the top-performing and most suitable one because the assessment is comprehensively done on time-series-based real-time data comparing the accuracy and F1 score values. Yet, it is noteworthy that in some of the cases, XGBoost and Gradient Boost demonstrate comparatively superior output.

TABLE I: Result Parameters for AAPL Stock

Model	Accuracy (%)	F1 Score (%)
SVM	99.53	99.72
Logistic Regression	99.06	99.45
Na¨ıve Bayes	91.98	95.15
Random Forest	98.58	99.16
Gradient Boost	98.73	99.18
XG Boost	99.25	99.56

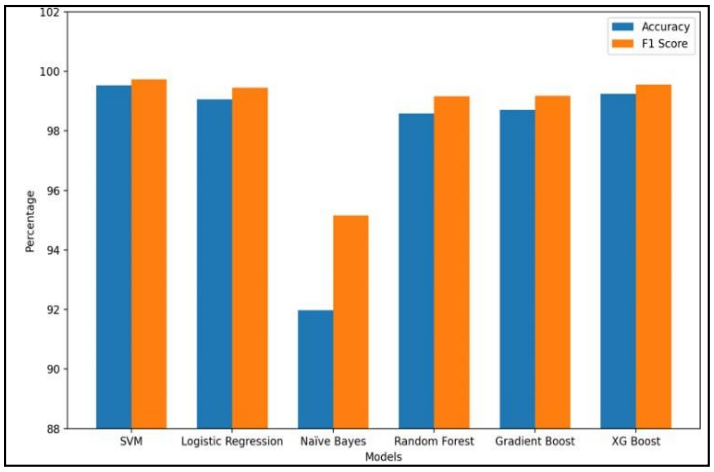

Fig. 4: Bar Plot of AAPL stock metrices

Table 1 illustrates the outcomes and Fig.4 demonstrates the bar plot obtained following the execution of ML models utilizing the data from the last two decades of Apple Inc.'s historical records. After observations, it can be deduced from the results that the best model that is chosen is SVM and the favorable output suggests that the stock is potentially buyable.

TABLE II: Result Parameters for BIOL Stock

Model	Accuracy (%)	F1 Score (%)
SVM	97.57	94.68
Logistic Regression	99.92	99.64
Na¨ıve Bayes	76.21	51.96
Random Forest	99.27	98.47
Gradient Boost	99.76	99.49
XG Boost	99.83	99.56

Table 2 and Fig.5 showcase the results obtained from ML model executions utilizing historical data spanning the past two decades for BIOLASE Inc. The model chosen here is Logistic Regression. As perceived from the outcomes, it is apparent that the output does not suggest the stock as being potentially buyable.

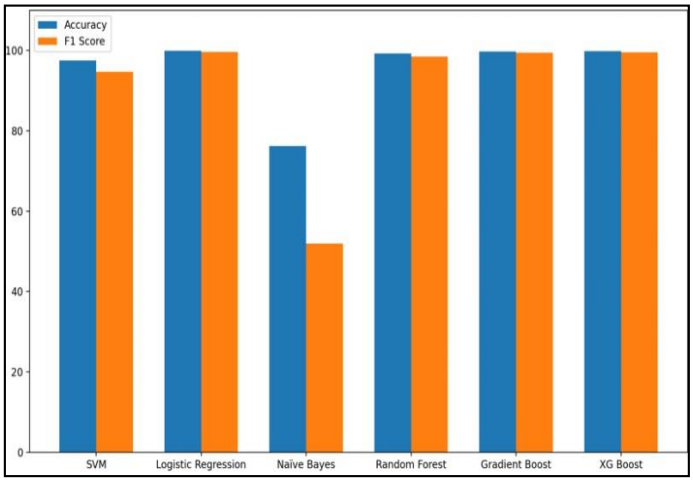

Fig. 5: Bar Plot of BIOL stock metrices

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VI. Conclusion and Future Work

In conclusion, this paper on time series-based stock market analysis by leveraging real-time stock data extraction through yfinance and applying machine learning algorithms (SVM, Logistic Regression, XGBoost, Na¨ıve Bayes, Random Forest, and Gradient Boost) has provided useful insights into anticipating stock movements. This comparative study utilizing accuracy and F1 score values highlighted the heterogeneity in algorithm performance across different stocks and conditions. In conclusion, the individual is advised on whether to proceed with purchasing the stock or to refrain from doing so. The results are real-time and dynamic because of which one cannot pertain that a specific algorithm is optimal. The choice is dynamically based on which model performs the best among all others. While it is evident that declaring a particular algorithm as optimal is difficult due to the volatility of real-time stock market data, it is worth noting that XGBoost and Gradient Boost tend to perform well across a wide range of conditions.

Moving forward, the future scope of this paper comprises the exploration of additional indicators to boost accuracy. The incorporation of deep learning techniques could significantly increase the models' potential to capture complex patterns in stock movements. Integrating user portfolios into the decision-making process could personalize suggestions, making them more aligned with individual investing goals. These enhancements seek to provide investors with more dependable and tailored insights, promoting improved decision-making in the ever-evolving field of stock market analysis.

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